

Boss Challenge — Paper 15 (Class 7)

Heat | Winds, Storms & Cyclones | Integers | Data Handling

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The degree of hotness or coldness of an object is what we call its:
(A) pressure
(B) density
(C) heat
(D) temperature
2. The sum of the integers (-5) and (+8) is:
(A) -13
(B) 3
(C) 13
(D) -3
3. Air that is moving from one place to another is what we simply call:
(A) rain
(B) pressure
(C) humidity
(D) wind
4. The arithmetic mean (average) of the numbers 4, 6 and 8 is:
(A) 4
(B) 6
(C) 18
(D) 9
5. The device specially designed to measure how hot or cold a body is:
(A) a speedometer
(B) a barometer
(C) a thermometer
(D) an anemometer
6. The value of $(-7) + (-4)$ is:
(A) -3
(B) -11
(C) 11
(D) 3
7. A key idea behind winds and storms is that air around us:
(A) only pushes downward
(B) cannot be felt ever
(C) has no weight at all
(D) exerts pressure

8. The mean of the values 10, 20 and 30 is:
- (A) 10
 - (B) 60
 - (C) 30
 - (D) 20
9. The temperature of a healthy human body is normally about:
- (A) 37°C
 - (B) 100°C
 - (C) 50°C
 - (D) 0°C
10. The value of $(-6) - (-9)$ is:
- (A) 3
 - (B) 15
 - (C) -3
 - (D) -15
11. Wind blows because air always moves from a region of high pressure toward a region of:
- (A) low pressure
 - (B) higher pressure
 - (C) the nearest ocean
 - (D) equal pressure everywhere
12. In the data set 2, 3, 3, 5 and 7, the mode is:
- (A) 4
 - (B) 3
 - (C) 7
 - (D) 5
13. A clinical thermometer is built to read only the small range needed for:
- (A) freezer temperatures below 0°C
 - (B) human body temperature (about 35°C to 42°C)
 - (C) water from ice to boiling (0°C to 100°C)
 - (D) furnace temperatures above 200°C
14. The product $(-4) \times (+5)$ equals:
- (A) 20
 - (B) 1
 - (C) -20
 - (D) -9
15. When air is warmed, it expands, becomes lighter and:
- (A) stops moving
 - (B) rises up
 - (C) turns into water
 - (D) sinks down

16. The median of the ordered data 3, 5, 7, 9 and 11 is:
- (A) 9
 - (B) 5
 - (C) 11
 - (D) 7
17. Heat always flows on its own from a body at higher temperature to one at:
- (A) a body of larger size
 - (B) the same temperature
 - (C) lower temperature
 - (D) higher temperature
18. The product $(-6) \times (-7)$ equals:
- (A) 13
 - (B) 42
 - (C) -42
 - (D) -13
19. The faster air moves over a place, the air pressure there becomes:
- (A) higher
 - (B) lower
 - (C) zero everywhere
 - (D) unchanged
20. The range of the data 12, 5, 9 and 20 is:
- (A) 15
 - (B) 25
 - (C) 5
 - (D) 20
21. The transfer of heat through a solid, from particle to particle without the solid moving, is:
- (A) convection
 - (B) evaporation
 - (C) conduction
 - (D) radiation
22. The value of $(-36) \div (+9)$ is:
- (A) -45
 - (B) -324
 - (C) -4
 - (D) 4
23. Winds are caused on a large scale mainly by the:
- (A) trees giving out air
 - (B) the rotation of the Moon
 - (C) uneven heating of the Earth by the Sun
 - (D) rivers flowing to the sea

24. The mean of 2, 4, 6, 8 and 10 is:

- (A) 10
- (B) 5
- (C) 30
- (D) 6

25. Materials such as metals, which let heat pass through them easily, are called:

- (A) conductors
- (B) insulators
- (C) radiators of light
- (D) non-metals

26. The value of $(-48) \div (-6)$ is:

- (A) 8
- (B) 8 with remainder
- (C) -54
- (D) -8

27. The seasonal winds that bring heavy rains to India each year are the:

- (A) trade-free winds
- (B) monsoon winds
- (C) land breezes
- (D) cyclonic winds

28. In a data set, the value that occurs most frequently is called the:

- (A) mean
- (B) median
- (C) mode
- (D) range

29. Materials like wood, plastic and wool, which do not let heat pass easily, are called:

- (A) liquids
- (B) conductors
- (C) insulators
- (D) metals

30. The additive inverse (opposite) of the integer -12 is:

- (A) 12
- (B) 0
- (C) -12
- (D) $1/12$

31. The instrument used to measure the speed of the wind is the:

- (A) anemometer
- (B) thermometer
- (C) barometer
- (D) rain gauge

- 32.** When data is arranged in order, the median is the:
- (A) largest value
 - (B) sum of all values
 - (C) most frequent value
 - (D) middle value
- 33.** The transfer of heat in liquids and gases by the actual movement of the heated fluid is:
- (A) conduction
 - (B) radiation
 - (C) convection
 - (D) freezing
- 34.** Among the integers -3, -7, 0 and -1, the greatest is:
- (A) -3
 - (B) 0
 - (C) -1
 - (D) -7
- 35.** A violent storm with very high-speed winds whirling around a centre of very low pressure is a:
- (A) a gentle sea breeze
 - (B) a monsoon shower
 - (C) cyclone
 - (D) a rainbow
- 36.** In a bar graph, the height (or length) of each bar represents the:
- (A) name of the item only
 - (B) value or frequency of that item
 - (C) colour chosen for the bar
 - (D) width of the bar
- 37.** Heat from the Sun reaches the Earth across empty space mainly by:
- (A) convection
 - (B) conduction
 - (C) radiation
 - (D) evaporation
- 38.** On the number line, the integer -4 lies to the ____ of -1:
- (A) above
 - (B) same spot as
 - (C) right
 - (D) left
- 39.** The calm, clear region at the very centre of a cyclone is known as the:
- (A) base of the cyclone
 - (B) shadow of the cyclone
 - (C) eye of the cyclone
 - (D) tail of the cyclone

40. A pictograph represents data using:
- (A) a single pie-shaped circle
 - (B) only long bars of equal width
 - (C) written paragraphs of text
 - (D) pictures or symbols, each standing for a number
41. The handle of a cooking pan is often made of plastic or wood because these are:
- (A) good conductors of heat
 - (B) heavier than metal
 - (C) better at reflecting light
 - (D) poor conductors of heat
42. The integer that is 5 more than -2 is:
- (A) -3
 - (B) -7
 - (C) 7
 - (D) 3
43. Lightning and thunder during a storm are caused by:
- (A) strong winds hitting trees
 - (B) a build-up of electric charges in the clouds
 - (C) the Sun shining through rain
 - (D) cold air freezing instantly
44. When a fair coin is tossed once, the probability of getting a head is:
- (A) 2
 - (B) 0
 - (C) 1
 - (D) $\frac{1}{2}$
45. Woollen clothes keep us warm in winter mainly because the wool fibres:
- (A) trap air, which is a poor conductor of heat
 - (B) themselves produce heat
 - (C) reflect sunlight onto the body
 - (D) are very good conductors
46. The integer that is 6 less than 2 is:
- (A) 8
 - (B) -8
 - (C) 4
 - (D) -4
47. During a thunderstorm, the safest thing to do is to:
- (A) swim in an open pool
 - (B) stay indoors, away from open fields and tall trees
 - (C) stand under the tallest tree
 - (D) fly a kite to watch the storm

48. When a fair six-faced die is rolled once, the probability of getting a 3 is:
- (A) $\frac{1}{3}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{1}{6}$
 - (D) $\frac{3}{6}$
49. During the day, the breeze near a coast usually blows from the sea to the land. This is called a:
- (A) land breeze
 - (B) monsoon wind
 - (C) cyclone
 - (D) sea breeze
50. A morning temperature of -5°C rises by 9 degrees by noon. The noon temperature is:
- (A) -4°C
 - (B) 14°C
 - (C) 4°C
 - (D) -14°C
51. In North America, the same kind of storm that India calls a cyclone is called a:
- (A) tornado
 - (B) monsoon
 - (C) hurricane
 - (D) blizzard
52. The probability of an event that is impossible is:
- (A) 100
 - (B) 1
 - (C) $\frac{1}{2}$
 - (D) 0
53. Why does a steel spoon feel colder than a wooden spoon, even when both are at room temperature?
- (A) steel conducts heat away from your hand faster
 - (B) steel is actually at a lower temperature
 - (C) steel reflects more cold
 - (D) wood gives out heat to your hand
54. A diver at -15 m (15 m below sea level) goes down another 8 m. Her new position is:
- (A) -15 m
 - (B) -7 m
 - (C) -23 m
 - (D) 23 m

55. A cyclone usually forms over warm seas because warm ocean water:
- (A) freezes into ice that spins
 - (B) evaporates and the rising moist air feeds the storm
 - (C) blocks all the wind
 - (D) turns into solid salt
56. The probability of an event that is certain to happen is:
- (A) $\frac{1}{2}$
 - (B) 1
 - (C) 2
 - (D) 0
57. The narrow bend (kink) in a clinical thermometer is there to:
- (A) show the room temperature too
 - (B) stop the mercury from falling back on its own so you can read it
 - (C) measure two patients at once
 - (D) make the mercury rise faster
58. The value of $(-1) \times (-1) \times (-1)$ is:
- (A) -1
 - (B) 1
 - (C) -3
 - (D) 3
59. If you blow hard between two hanging paper strips, they move toward each other because:
- (A) your breath is warmer than the room
 - (B) the paper attracts itself like a magnet
 - (C) fast-moving air between them lowers the pressure there
 - (D) the air pushes them apart
60. Over a week, the daytime temperatures were 30° , 32° , 31° , 33° , 29° , 34° and 31°C . Their mean is:
- (A) 7°C
 - (B) 34°C
 - (C) 31°C
 - (D) 217°C
61. In hot summer weather, light-coloured clothes are more comfortable because light colours:
- (A) conduct heat to the skin faster
 - (B) absorb more heat and stay hot
 - (C) produce a cooling breeze
 - (D) reflect more heat and absorb less

62. The product of three negative integers is always:
- (A) always 1
 - (B) positive
 - (C) negative
 - (D) zero
63. Cyclone warnings to coastal communities are issued by the:
- (A) the electricity board
 - (B) meteorological (weather) department
 - (C) the local post office
 - (D) the railway station
64. A weather station logs wind speeds of 12, 15, 18 and 15 km/h on four days. The mode is:
- (A) 60 km/h
 - (B) 12 km/h
 - (C) 18 km/h
 - (D) 15 km/h
65. A small gap is deliberately left between two lengths of railway track because metal:
- (A) shrinks when heated and pulls apart
 - (B) must be cleaned of dust regularly
 - (C) expands when it gets hot and needs room to grow
 - (D) carries electricity better with a gap
66. The value of $0 \times (-15)$ is:
- (A) 0
 - (B) 15
 - (C) -15
 - (D) -0 meaning less than zero
67. Which action is a sensible part of preparing for an approaching cyclone?
- (A) store drinking water, food and move valuables to a safe, high place
 - (B) open all windows to let the wind through
 - (C) go out to the beach to watch the waves
 - (D) ignore the warning and carry on as usual
68. When a data set has an even number of ordered values, the median is found by taking:
- (A) the average of the two middle values
 - (B) the first value in the list
 - (C) the largest of all the values
 - (D) the sum of all the values

69. When the bottom of a beaker of water is heated, the warm water rises because on heating it becomes:

- (A) lighter (less dense)
- (B) solid like ice
- (C) colder than before
- (D) heavier (more dense)

70. The only integer lying strictly between -2 and 0 is:

- (A) 0
- (B) 1
- (C) -1
- (D) -2

71. By day, the land near a coast heats up faster than the sea because land:

- (A) warms up more quickly than water for the same sunshine
- (B) water warms up faster than land
- (C) the sea blocks all sunlight
- (D) land is closer to the Sun

72. If the mean of 4 numbers is 10, then the total (sum) of those numbers is:

- (A) 14
- (B) 40
- (C) 10
- (D) 2.5

73. A laboratory thermometer differs from a clinical one mainly because it can measure:

- (A) wind speed in a storm
- (B) a much wider range of temperatures
- (C) only the human body's temperature
- (D) the weight of a liquid

74. The sum of any integer and its additive inverse is always:

- (A) the number itself
- (B) 0
- (C) 1
- (D) always positive

75. Strong cyclonic winds combined with heavy rain and high sea waves cause the most damage to:

- (A) low-lying coastal areas
- (B) underground caves
- (C) the middle of a desert
- (D) high mountain peaks far inland

76. A double bar graph is especially useful when you want to:
- (A) display fractions of a whole
 - (B) show a single set of data only
 - (C) compare two sets of data side by side
 - (D) record data without any numbers
77. The common everyday unit used to express temperature in India is the:
- (A) degree Celsius ($^{\circ}\text{C}$)
 - (B) kilogram
 - (C) litre
 - (D) metre
78. The value of $(-8) + 8 + (-3)$ is:
- (A) -19
 - (B) 13
 - (C) 3
 - (D) -3
79. Increasing the speed of wind over a region tends to make objects there experience:
- (A) an upward or inward push as pressure drops
 - (B) no force at all
 - (C) a stronger downward pull from the wind
 - (D) a rise in their temperature
80. The range of a data set is calculated as:
- (A) highest value plus lowest value
 - (B) highest value minus lowest value
 - (C) the most frequent value
 - (D) total divided by count
81. A thermos flask keeps drinks hot or cold for long because its design mainly tries to:
- (A) conduct heat quickly through metal
 - (B) speed up heat transfer to the air
 - (C) turn the liquid into a gas
 - (D) reduce heat transfer by conduction, convection and radiation
82. A bank balance of -200 (overdrawn) after a deposit of 500 becomes:
- (A) 700
 - (B) -700
 - (C) -300
 - (D) 300
83. Which statement about air pressure and wind is correct?
- (A) wind blows only where pressure is equal
 - (B) wind always blows from low to high pressure
 - (C) air pressure has nothing to do with wind
 - (D) wind is caused by differences in air pressure between two places

84. The mean of the numbers 0, 0, 0 and 4 is:

- (A) 0
- (B) 1
- (C) 4
- (D) 2

85. On a sunny day, a person feels warmth from sunlight even with cool air around, showing that radiation:

- (A) needs a metal wire to travel
- (B) can transfer heat without warming the air in between
- (C) must heat all the air first
- (D) only works by touching the Sun

86. The property shown by $a + b = b + a$ for all integers a and b is the:

- (A) distributive property
- (B) closure property
- (C) commutative property of addition
- (D) associative property

87. A simple home experiment crushing a sealed empty tin after heating and cooling shows that:

- (A) heat makes metal weak forever
- (B) air has no pushing force
- (C) water inside expands and bursts it
- (D) air pressure can push hard enough to crush the tin

88. The mean of the integers -2, 0 and 2 is:

- (A) 4
- (B) -2
- (C) 2
- (D) 0

89. Black or dark-coloured surfaces are better than light ones at:

- (A) absorbing heat
- (B) staying coolest in the Sun
- (C) blocking all sunlight
- (D) reflecting heat

90. The value of $(-2) \times 3 \times (-5)$ is:

- (A) -4
- (B) 30 with a remainder
- (C) 30
- (D) -30

- 91.** Thunderstorms are most likely to develop in weather that is:
- (A) cold and dry
 - (B) calm and freezing
 - (C) dusty with no clouds
 - (D) hot and humid
- 92.** When a fair six-faced die is rolled, the probability of getting an even number is:
- (A) $\frac{1}{2}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{1}{6}$
 - (D) 1
- 93.** If a thermometer reading rises from 28°C in the morning to 36°C by noon, the rise in temperature is:
- (A) 64°C
 - (B) 28°C
 - (C) 8°C
 - (D) 4°C
- 94.** The difference between a temperature of 6°C and one of -4°C is:
- (A) 10°C
 - (B) -10°C
 - (C) 24°C
 - (D) 2°C
- 95.** The main reason people are advised to move to a cyclone shelter is that the shelter is:
- (A) the only place with mobile signal
 - (B) where the storm never reaches
 - (C) a place to watch the storm closely
 - (D) a strong building that can withstand high winds and flooding
- 96.** In the marks 5, 5, 6, 7 and 8, the mode is:
- (A) 31
 - (B) 5
 - (C) 8
 - (D) 6
- 97.** A clinical thermometer should never be washed in very hot water because:
- (A) the mercury may expand too much and break the thermometer
 - (B) hot water makes it read too low
 - (C) it will then only read room temperature
 - (D) the numbers will wash off

98. Writing -1, -5, 3 and -2 in descending (largest to smallest) order gives:

- (A) 3, -1, -2, -5
- (B) -5, -2, -1, 3
- (C) -1, -2, 3, -5
- (D) 3, -5, -2, -1

99. The simple device, often shaped like an arrow on a roof, that shows the direction from which the wind blows is a:

- (A) anemometer
- (B) wind vane
- (C) rain gauge
- (D) thermometer

100. A pie chart (circle graph) is best used to show:

- (A) how a whole is divided into parts
- (B) the exact wind speed each hour
- (C) the precise temperature of a body
- (D) two data sets side by side